Software Engineering Lab 1

Problem Statement #53 – Freelance Content Creator Escrow Platform

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow creators to submit draft media deliverables for client review and release locked milestone payments upon sponsor approval sign-off.	High	Pass: A submitted milestone can be reviewed by the sponsor, and its locked payment is released after sponsor approval. Fail: Funds are released without client sign-off.	This is the core workflow connecting deliverable submission, approval, and milestone payment.
FR-002	Functional	The system shall automatically apply a watermark to uploaded image and video draft deliverables before they are made available for client review.	High	Pass: Every uploaded image/video draft presented for client review contains the required watermark. Fail: An unwatermarked draft is made available for review.	Protects the creator's draft media before client approval.
FR-003	Functional	The system shall allow the client sponsor to view and review a creator's submitted watermarked draft for the corresponding milestone.	High	Pass: The sponsor can open the milestone and view the submitted watermarked draft. Fail: The sponsor cannot access or review the submitted draft.	Client review is necessary before a milestone can be approved.
FR-004	Functional	The system shall allow the client sponsor to approve or reject a submitted milestone deliverable.	High	Pass: The sponsor can select either Approve or Reject, and the system records the selected decision against the milestone. Fail: The system does not record the sponsor's decision.	Provides the sponsor with formal control over milestone acceptance.
FR-005	Functional	The system shall release the locked milestone payment to the creator only after the corresponding milestone has received sponsor approval.	High	Pass: Payment is released after valid sponsor approval. Fail: Payment is released when the milestone has not been approved.	Prevents unauthorized payment release and supports the escrow workflow.
NFR-001	Nonfunctional — Performance & Security	Uploaded media drafts must process automated watermarking in under 5 seconds for video/image previews.	High	Pass: Benchmarking confirms that watermarking completes within 5 seconds under the specified test conditions. Fail: Watermarking exceeds 5 seconds.	Ensures that watermarking does not significantly delay the client's review process.
NFR-002	Nonfunctional — Security	The system shall prevent milestone funds from being released unless valid client sponsor approval has been recorded for the corresponding milestone.	High	Pass: In all security tests where sponsor approval is absent, milestone funds remain locked. Fail: Funds are released without recorded sponsor approval.	Protects escrowed funds and ensures that payment release follows the required approval process.

Software Engineering Lab 1 – UML Use-Case Diagram

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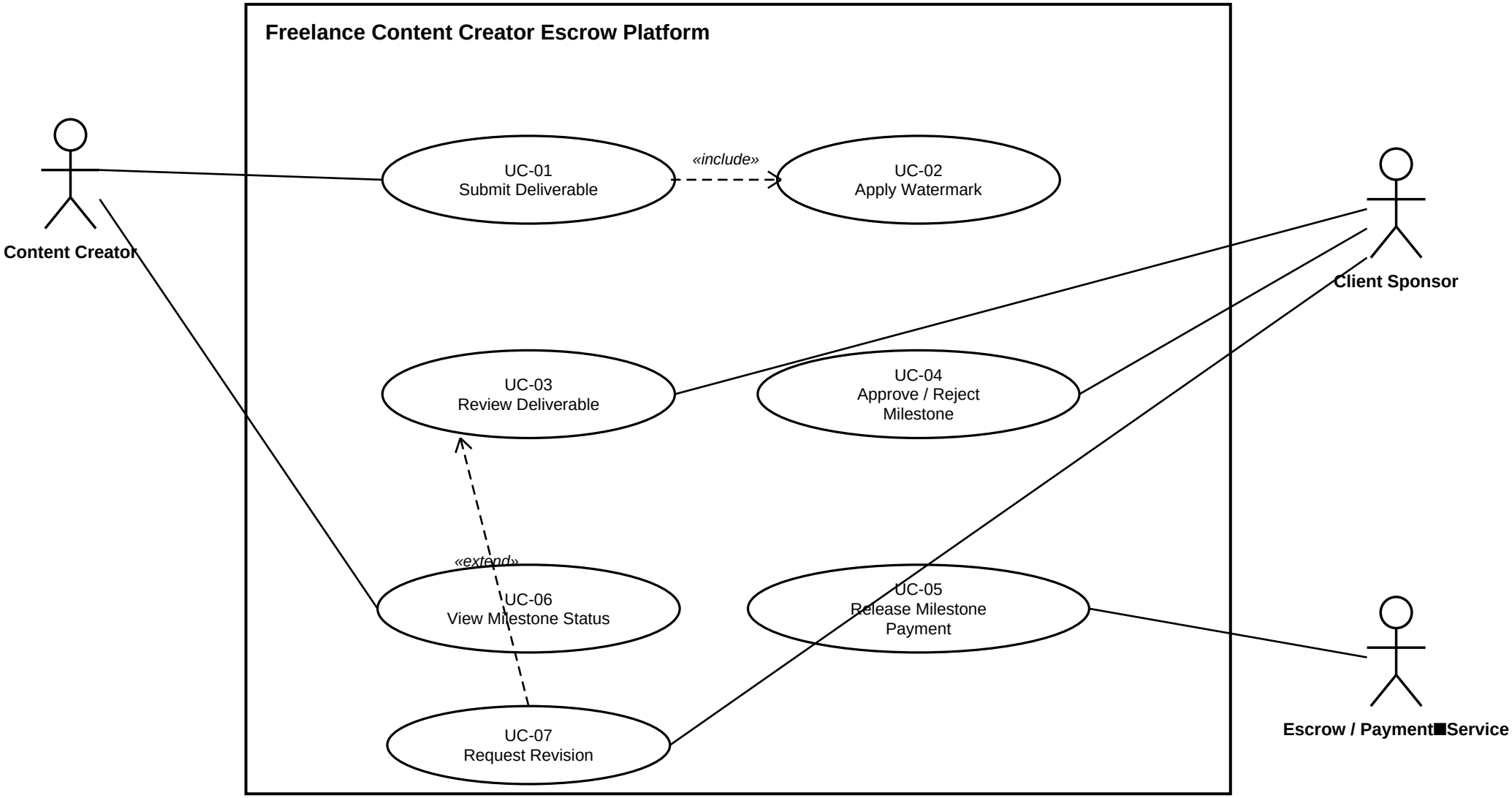

Legend: Dashed arrow «include» = mandatory reused behavior; «extend» = optional/conditional behavior.

Software Engineering Lab 1 – Use-Case Flow Specification

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Use Case: Approve Milestone and Release Payment

Actors: Client Sponsor, Content Creator, Escrow / Payment Service

Preconditions

1. A contract milestone exists between the Content Creator and Client Sponsor.
2. The Content Creator has submitted the required milestone deliverable.
3. The submitted media draft has been watermarked and is available for sponsor review.
4. The corresponding milestone payment is locked in escrow.

Main Success Scenario

1. The Client Sponsor opens the submitted milestone deliverable.
2. The system displays the watermarked draft for review.
3. The Client Sponsor reviews the deliverable against the milestone requirements.
4. The Client Sponsor selects **Approve**.
5. The system records the sponsor's approval for the milestone.
6. The system marks the milestone as completed.
7. The system sends a payment-release request to the Escrow / Payment Service.
8. The Escrow / Payment Service releases the locked milestone payment to the Content Creator.
9. The system records the payment release and displays confirmation to the relevant parties.
10. The use case ends successfully.

Postconditions

• The milestone is marked as completed.
• The Client Sponsor's approval is recorded.
• The locked milestone payment has been released to the Content Creator.
• The approval and payment-release transaction is recorded by the system.

Alternate Flow – Deliverable Rejected

A1. At Step 4, the Client Sponsor selects **Reject** instead of Approve.
A2. The system records the rejection and requests revision feedback.
A3. The Client Sponsor provides feedback describing the required changes.
A4. The system sends the feedback to the Content Creator and keeps the milestone incomplete.
A5. The milestone payment remains locked, and the Content Creator may submit a revised deliverable.
A6. The use case ends without releasing the payment.